# RELIABILITY APPLIED TO SLOPE STABILITY ANALYSIS

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**ABSTRACT:** Formally probabilistic methods for the analysis of slope stability have had relatively little impact on practice. Many engineers are not familiar with probabilistic concepts, and it has been difficult to incorporate concepts of reliability into practice. Also, there is confusion over what reliability and probability of failure mean. The most effective applications of probabilistic methods are those involving relative probabilities of failure or illuminating the effects of uncertainties in the parameters. Attempts to determine the absolute probability of failure are much less successful. The paper describes how probabilistic descriptions of soil parameters can be derived from field and laboratory data and applied in stability analysis. The first-order, second-moment approach is explored and applied to the design of embankment dams. The example illustrates the relative contributions of uncertainties about different parameters to the reliability of the embankment. Reliability analysis is especially useful in establishing design values of factors of safety representing consistent risks for different types of failure.

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## INTRODUCTION

Geotechnical engineers deal with uncertainties by recognizing that risk and uncertainty are inevitable and by applying the observational method (Peck 1969) to maintain control over them. However, the observational method is applicable only when the design can be changed during construction on the basis of observed behavior. In those cases in which the critical behavior cannot be observed until too late to make changes, the designer must rely on a calculated risk. Although prominent engineers have long emphasized the role of the calculated risk in geotechnical engineering (Casagrande 1965), the literature contains relatively few examples of formal evaluation of risk for slopes and embankments, and the profession has been relatively less vigorous in the use of modern probabilistic techniques than have other branches of engineering.

The most effective applications of probabilistic concepts to analysis of the stability of slopes and embankments have used the so-called first-order, second-moment methods described by Wu and Kraft (1970), Cornell (1971), Alonso (1976), Tang et al. (1976), and Vanmarcke (1977), among others. The techniques have now evolved from purely theoretical investigations into methods that can be applied to practical problems. In recent years there have been a number of applications to slope stability (e.g., Li and Lumb 1987, Luckman et al. 1987; Halim and Tang 1991). Nevertheless, the application has not been as widespread as would be warranted by their power or by the original developers' expectations.

The present paper, based on an earlier presentation (Christian et al. 1992) at a conference devoted to developments in the understanding of the behavior of slopes and embankments, addresses how probabilistic methods

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should be used and for what ends. It first reviews some basic, well-known ideas about probabilistically based reliability. It then describes the application of reliability methods to the evaluation of slope stability, primarily by means of a specific example. Finally, it summarizes some of the lessons that have been distilled from such practical applications.

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## SCOPE OF PROBABILISTIC ANALYSIS

What is meant by the "probability of failure"? The established methods for performing probabilistic slope stability analysis deal primarily with uncertainty in the analyst's knowledge of material properties and loading conditions. They do not address inadequacies in spillway capacity, failure to prevent or detect internal erosion, or errors in construction, and they seldom consider that an inadequate field exploration program might miss a critical geological detail, even though these are among the major causes of failure of slopes and embankments. Indeed, Baecher et al. (1980) determined that earth dams seem to have probabilities of failure of about $10^{-4}$ per dam-year for a very wide range of locations and times of construction. Clearly any probability of failure that is to be used in benefit-cost analysis or probabilistic risk assessment must be based on more than the uncertainty in known geotechnical properties.

Even if the calculated probability of failure were to include many factors that fall outside of the usual scope of stability analysis, it would still not be clear how to use the resulting numbers. Members of the public, and for that matter most engineers, have great difficulty understanding the implications of events with large consequences and small probabilities of occurrence. People readily accept the relatively large risks of such activities as driving cars but shun unfamiliar activities of lower risk. It is well known that acceptance of risk is not purely rational, and simply providing numbers for annual probability of failure of an engineered structure without some evaluation is an invitation to misunderstanding and confusion.

It is far more useful to deal with relative probabilities of failure or other measures of reliability. The engineer can then incorporate uncertainty into the decision-making process without trying to establish an absolute value of the probability of failure. This is the approach used here. It should be noted that the techniques used in this paper are commonly referred to as the mean-first-order reliability method (MFORM), which is a simplification of the more general first-order reliability method (FORM).

## BASIC CONCEPTS

Some definitions and notational conventions need to be established before proceeding further. If the two properties or variates in question are $x$ and $y$, and they are sampled at $n$ points identified by the subscript $i$, then the following definitions apply. (It is understood that $\Sigma$ implies summation over all possible values.)

The mean or expected value, $E[x]$, is

$$E[x] = (1/n) \sum x_i \tag{1}$$

The variance, $V[x]$, is

$$V[x] = [1/(n - 1)] \sum (x_i - E[x])^2 \tag{2}$$

The standard deviation, $\sigma[x]$, is

$$\sigma[x] = (V[x])^{1/2} \tag{3}$$

The coefficient of variation, Cov($x$), is

$$\text{Cov}[x] = \frac{\sigma[x]}{E[x]} \tag{4}$$

Because the total variance due to several independent variates is the sum of the individual variances, coefficients of variation and standard deviations are combined by taking the square root of the sum of the squares of the individual components.

The covariance of two variates, $C[x, y]$, is

$$C[x, y] = [1/(n - 1)] \sum (x - E[x]) \cdot (y - E[y]) \tag{5}$$

The autocovariance of a single variate over a distance $r$, $C_x(r)$, is

$$C_x(r) = [1/(n - 1)] \sum (x_i - E[x]) \cdot (x_{i+r} - E[x]) \tag{6}$$

The autocorrelation coefficient, $R_x(r)$, is

$$R_x(r) = C_x(r)/V[x] \tag{7}$$

The autocovariance and autocorrelation describe how well the value of a parameter at a given point can predict the value at another point some distance away. Because these are both functions of distance $r$, it is often convenient to define the autocorrelation distance, $r_0$, as the distance at which $R_x(r)$ due to spatial variability decreases to 37% (the inverse of the base of natural logarithms, $e$) of the value at $r = 0$. This provides a single number with which to describe the smoothness of spatial variability. A large autocorrelation distance implies a very uniform material, and small distance implies a material whose properties change over short distances.

The most effective applications of probability theory to the analysis of slope stability have stated the uncertainties in the form of a reliability index, which is defined as

$$\beta = \frac{E[F] - 1.0}{\sigma[F]} \tag{8}$$

In (8), $\beta$ = reliability index, and $F$ = computed factor of safety. The reliability index describes safety by the number of standard deviations separating the best estimate of $F$ from its defined failure value of 1.0. It can also be considered as a way of normalizing the factor of safety with respect to its standard deviation. The definition does not require that the probability distribution of $F$ be known, but there is a tacit assumption that $F$ is distributed in some way that can be described meaningfully by an expected value and standard deviation.

When the shape of the probability distribution of the factor of safety is known, the reliability index can be related to the probability of failure. Fig. 1 shows the relation for a normally distributed factor of safety. Other distributions give similar curves for values of $\beta$ less than 1.5 to 2.0, but the curves differ substantially for higher values of $\beta$. For any consistently made assumption on the probability distribution of $F$, $\beta$ is monotonically related to the probability of failure, so higher values of $\beta$ imply lower probabilities of failure. Of course, the probability of failure incorporates only those factors that are included in the analysis and excludes the many important



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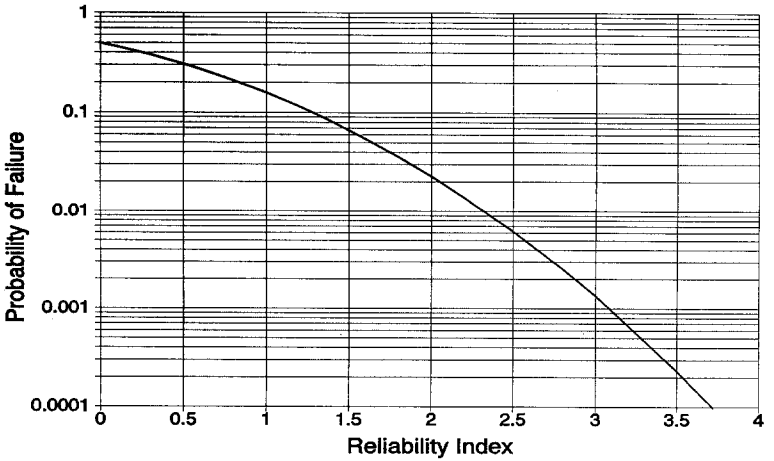



**FIG. 1. Nominal Probability of Failure for Normally Distributed $F$ as Function of Reliability Index**

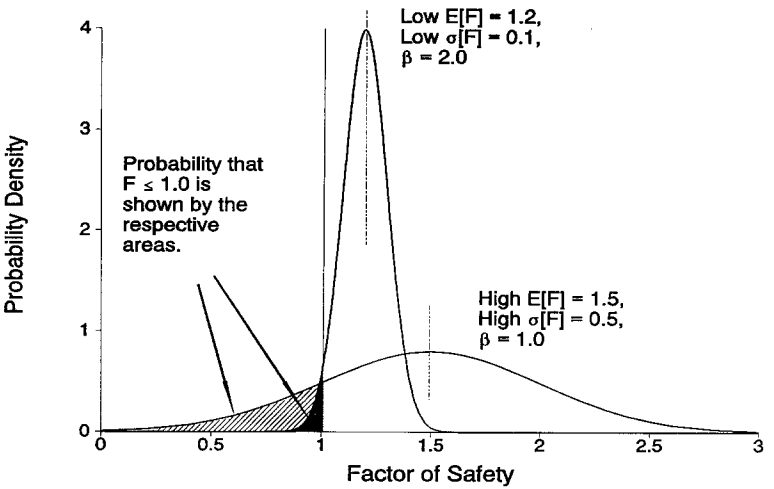



**FIG. 2. Comparison of Two Situations with Different Means and Distributions of Factor of Safety**

factors mentioned in the previous section. Therefore, it must be considered a lower bound on the actual probability of failure.

Whether or not the probability distribution of $F$ is known, the reliability index expresses uncertainty in the stability of the slope. Fig. 2 compares two idealized cases, in both of which $F$ is assumed to be normally distributed about the mean value. It is clear that the case with the mean $F$ of 1.5 and $\sigma[F]$ of 0.5 has a higher overall probability of failure than does the case with a mean $F$ of 1.2 and a smaller $\sigma[F]$ of 0.1. The rational approach is to develop a design to achieve a consistent reliability index rather than relying on a value of $F$ alone. This should provide a more comprehensive

accounting of the relevant information, for one does, in fact, know more about the stability of the slope than $F$ alone admits.

The reliability index can also be expressed in terms of the coefficient of variation of $F$, defined in (4)

$$\beta = \frac{E[F] - 1.0}{E[F] \cdot \text{Cov}[F]} \tag{9}$$

From (9), it is easy to select design values of $F$ that have the same reliability index. For example, to achieve $\beta$ of 2, when Cov[$F$] is 0.2, the computed $F$ must be 1.67, but, when Cov[$F$] is 0.1, the computed $F$ need be only 1.25. Thus, the reliability index expressed in terms of the coefficient of variation of the factor of safety provides an internally consistent criterion for design. How then is the variation of $F$ to be found?

## FIRST-ORDER PREDICTIVE MODEL TO ESTIMATE UNCERTAINTIES

The predictive model assumes that a performance variable such as the factor of safety $F$ can be described by

$$F = g(\mathbf{x}) + e \tag{10}$$

where $g$ = a geotechnical model such as the simplified Bishop method of slices that depends on a vector $\mathbf{x}$ of geometry and soil properties, and $e$ = modeling error that describes how the model deviates from the real world. The uncertainty in $F$ can be measured by its variances, and these can be estimated by expanding (10) in a Taylor series and truncating to the first terms to yield

$$V[F] \approx \sum_{i=1}^{k} \sum_{j=1}^{k} \frac{\partial g}{\partial x_i} \frac{\partial g}{\partial x_j} C[x_i, x_j] + V[e] \tag{11}$$

where $C[x_i, x_j]$ = covariance of the parameters $x_i$ and $x_j$ contributing to the calculations of $F$ and

$$C[x_i, x_i] = V[x_i] \tag{12}$$

If all the $x_i$ and $x_j$ are uncorrelated for $i \neq j$, then

$$V[F] \approx \sum_{i=1}^{k} \left(\frac{\partial g}{\partial x_i}\right)^2 V[x_i] + V[e] \tag{13}$$

making the simplifying assumption that uncertainty in the model error $e$ is uncorrelated with the uncertainties in the model parameters $x_i$ and $x_j$. These equations use the variances, or second moments, of the distributions, and they are related by first-order approximations. Hence, the approach is called a first-order, second-moment method.

There are two direct ways of evaluating (11) or (13). First, the model, $g$, can be differentiated formally to give a closed form expression for $V[F]$. However, in most practical cases the computation of the closed-form derivatives is inconvenient or impossible, and a second approach is needed. This is to calculate divided differences as approximations to the partial derivatives.



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### Computing Uncertainty in Factor of Safety and Calculating Reliability Index

The computation of factor of safety using the method of slices for an actual geometry is an example of a case in which direct differentiation is not feasible, and the second method of evaluating the variance must be used. The recommended approach for computing the uncertainty in the factor of safety and the corresponding reliability index can be divided into the following six steps:

1. Identify all variables ($x_i$) that affect the stability of the particular embankment. Prime examples are the geometry, weight and strength of fill materials, and the relevant properties of the foundation soils.
2. Determine the best estimate (usually the mean value) of each variable, $E[x_i]$, and use these to calculate the best estimate of the factor of safety, $E[F]$, with a method of slices that satisfies both force and moment equilibrium.
3. Estimate the uncertainty in each of the variables and, in particular, its likely variance, $V[x_i]$. For example, the next section discusses uncertainty in soil properties.
4. Perform sensitivity analyses by calculating the changes in factor of safety due to changing each variable and computing $\Delta F/\Delta x_i$.
5. Use (13) to obtain $V[F]$.
6. Use (8) to calculate the reliability index, where $\sigma[F]$ is the square root of $V[F]$.

The numbers in Tables 1–4 are the results of the first three steps. Tables

**TABLE 1. Soil Profile and Fill Property Uncertainties for Reliability Analysis**

| Variable (1) | Expected value (2) | Variance | | |
|---|---|---|---|---|
| | | Spatial (3) | Systematic (4) | Total (5) |
| Depth of crust | 4 m | 0.96 | 0.04 | 1.0 |
| Depth to till | 18.5 m | 0.0 | 1.0 | 1.0 |
| Fill density | 20 kN/m$^3$ | 1.0 | 1.0 | 2.0 |
| Fill friction $\phi'$ | 30° | 1.0 | 3.0 | 4.0 |

**TABLE 2. Undrained Shear Strength Uncertainty Based on Field Vane Data**

| Field vane statistics (1) | Clay | |
|---|---|---|
| | Marine (2) | Lacustrine (3) |
| Number of tests | 62 | 37 |
| Mean (kPa) | 34.5 | 31.2 |
| Data scatter (Cov) | 0.236 | 0.272 |
| Spatial variability (Cov) | 0.183 | 0.272 |
| Systematic error (Cov) | | |
| Statistical | 0.030 | 0.045 |
| Correction factor ($\mu$) | 0.075 | 0.15 |
| Subtotal | 0.08 | 0.16 |
| Total Cov | 0.20 | 0.32 |

**TABLE 3. Undrained Strength Ratios from Laboratory $CK_0U$ Tests**

| Deposit (1) | State (2) | $\tau_C/\sigma'$ (3) | $\tau_D/\sigma'$ (4) | $\tau_E/\sigma'$ (5) |
|---|---|---|---|---|
| B-6 Marine | OC | 0.26 | 0.225 | 0.16 |
| B-6 Marine | NC | 0.26 | 0.225 | 0.16 |
| B-6 Lacustrine | OC | 0.225 | 0.19 | 0.14 |
| B-6 Lacustrine | NC | 0.25 | 0.215 | 0.12 |

Note: $\sigma' = \sigma'_p$ for OC clay; and $\sigma' = \sigma'_{vc}$ for NC clay.

**TABLE 4. Uncertainty in Undrained Shear Strength Estimates for Multistage Analysis**

| Parameter (1) | Clay (2) | Coefficient of Variation | | |
|---|---|---|---|---|
| | | Spatial (3) | Systematic (4) | Total (5) |
| | | (*a*) Intact Clay | | |
| $\tau/\sigma'_p$ | Marine | 0.067 | 0.076 | 0.10 |
| $\tau/\sigma'_p$ | Lacustrine | 0.084 | 0.092 | 0.12 |
| $\sigma'_p$ | Marine | 0.18 | 0.10 | 0.21 |
| $\sigma'_p$ | Lacustrine | 0.27 | 0.15 | 0.31 |
| $\tau$ | Marine | 0.19 | 0.13 | 0.23 |
| $\tau$ | Lacustrine | 0.28 | 0.18 | 0.33 |
| | | (*b*) Consolidated Clay | | |
| $\tau/\sigma'_{vc}$ | Marine | 0.067 | 0.076 | 0.10 |
| $\tau/\sigma'_{vc}$ | Lacustrine | 0.084 | 0.092 | 0.12 |
| $\sigma'_{vc}$ | Marine | 0.09 | 0.08 | 0.12 |
| $\sigma'_{vc}$ | Lacustrine | 0.09 | 0.08 | 0.12 |
| $\tau$ | Marine | 0.11 | 0.11 | 0.16 |
| $\tau$ | Lacustrine | 0.12 | 0.12 | 0.17 |

5 and 6 show the results of the last three steps for two cases. These are now discussed in more detail.

## UNCERTAINTY IN SOIL PROPERTIES

The uncertainty in the stability of a slope is the result of many factors. Some, such as ignorance of geological details missed in the exploration program, are difficult to treat formally; others, such as the estimates of soil properties are more amenable to statistical analysis.

Fig. 3 illustrates how the uncertainties in soil properties can arise from two sources: scatter in the data and systematic error in the estimate of the properties. The former consists of inherent spatial variability in the properties and random testing errors in their measurement. The latter consists of systematic statistical errors due to the sampling process and bias in the measurement process itself. In summary, selecting design shear strength presents four problems:

1. Scatter in the data due to real spatial variability within the profile
2. Scatter in the data due to random testing errors or noise
3. A systematic error in the computed mean value of the property due to the limited number of tests performed, leading to a statistical uncertainty



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**TABLE 5. Variance Composition, Single-Stage Dike, $H$ = 12 m**

| Parameter (1) | $\Delta F/\Delta x_i$ (2) | Variance: Systematic (3) | Variance: Total (4) | $(\Delta F/\Delta x_i)^2 \cdot V(x_i)$: Systematic (5) | $(\Delta F/\Delta x_i)^2 \cdot V(x_i)$: Total (6) |
|---|---|---|---|---|---|
| $\phi'$ | 0.01 | 3.0 | 4.0 | 0.0003 | 0.0004 |
| $\gamma_{fill}$ | 0.06 | 1.0 | 2.0 | 0.0036 | 0.0072 |
| $D_{crust}$ | 0.008 | 0.04 | 1.0 | nil | 0.0001 |
| $D_{till}$ | 0.056 | 1.0 | 1.0 | 0.0031 | 0.0031 |
| $c_u(M)$ | 0.0137 | 7.6 | 47.6 | 0.0014 | 0.0089 |
| $c_u(L)$ | 0.0215 | 24.9 | 99.7 | 0.0115 | 0.0461 |
| | | | $V[F]$ | 0.0199 | 0.0658 |
| | | | Spatial $V[F]$ | 0.0459 | |
| | | | $V_{ave}[F]$ at $R$ = 0.2 | 0.0199 + 0.0092 | |
| | | | | 0.0291 | |
| | | | $\beta$ | $(1.453 - 1.0)/\sqrt{0.0219}$ | |
| | | | | 2.66 | |

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**TABLE 6. Variance Composition, Multistage Dike, $H$ = 23 m**

| Parameter (1) | $\Delta F/\Delta x_i$ (2) | Variance | | $(\Delta F/\Delta x_i)^2 \cdot V(x_i)$ | |
|---|---|---|---|---|---|
| | | Systematic (3) | Total (4) | Systematic (5) | Total (6) |
| $\phi'$ | 0.0088 | 3.0 | 4.0 | 0.0002 | 0.0003 |
| $\gamma_{fill}$ | 0.055 | 1.0 | 2.0 | 0.0030 | 0.0061 |
| (*a*) Intact Clay | | | | | |
| $\tau(M)$ | 0.0018 | 13.3 | 41.5 | nil | 0.0001 |
| $\tau(L)$ | 0.012 | 26.3 | 88.5 | 0.0038 | 0.0127 |
| (*b*) Consolidated Clay | | | | | |
| $\tau(M)$ | 0.0021 | 52.7 | 111.5 | 0.0002 | 0.0005 |
| $\tau(L)$ | 0.009 | 62.0 | 124.4 | 0.0050 | 0.0101 |
| | | | $V[F]$ | 0.0122 | 0.0298 |
| | | | Spatial $V[F]$ | 0.0176 | |
| | | | $V_{ave}[F]$ at $R$ = 0.07 | 0.0122 + 0.0012 | |
| | | | | 0.0134 | |
| | | | $\beta$ | $(1.427 - 1.0)/\sqrt{0.0134}$ | |
| | | | | 3.69 | |

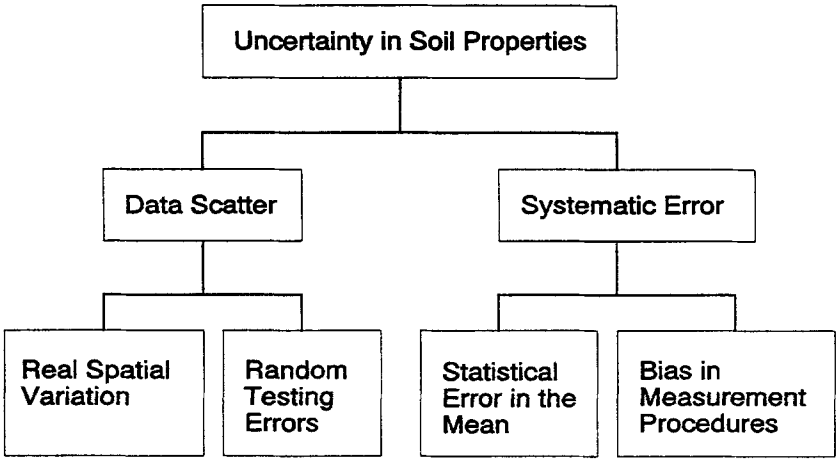



**FIG. 3. Categories of Uncertainty in Soil Properties**

4. An error in the mean due to bias in the measurement, such as those known to occur in field vane testing or from the effects of excessive sample disturbance.

## ASSESSING SOIL PROPERTIES AT JAMES BAY

The application of probabilistic and statistical methods is illustrated by work done by the last two authors as part of activities to recommend techniques for designing embankment dams on soft sensitive clays for the James Bay hydroelectric project in northern Quebec (Ladd et al. 1983; Ladd 1991). The James Bay hydroelectric project required the construction of approximately 50 km of dikes on soft, sensitive clays. When the stability of embankments in such circumstances is evaluated, professional opinions often diverge regarding the strengthening of foundation soil during construction, selection of appropriate factors of safety, and evaluation of strength properties. An international committee was formed to address and resolve the issues of stability of the slopes. The goals of the analysis included developing and understanding the relative safety of different designs, obtaining insight about the influence of different parameters, and establishing consistent design criteria that could be used in preliminary designs. One of the first matters to be dealt with was the estimation of the initial strength of the in-situ foundation soils.

### Data Scatter: Spatial Variability and Noise

Field vane tests conducted with the Nilcon device on the James Bay clays exhibited an unusually large degree of scatter. If $\mu$ is Bjerrum's (1972) empirical factor for correcting the bias in field vane data, then a common model of measurement error is

$$\mu c_u(FV) = c_u + c_e \tag{14}$$

where $c_u$ = actual undrained strength; $c_u(FV)$ indicates the value measured by the field vane test; and $c_e$ = a random experimental error with a zero mean. Eq. (14) assumes that $\mu$ is known precisely. The engineer needs to



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know the actual in-situ shear strength, $c_u$, including its spatial variability, but the random experimental variations represented by $c_e$ should be eliminated. One way to do this is to use the autocovariance function, provided it does not change with position.

Fig. 4 illustrates the analysis. First the expected value—the mean or the trend in the data—must be evaluated by some appropriate method such as linear regression. Then the autocovariance is a measure of the joint variation of the property from the expected value as a function of the distance between the points where the property is measured. Fig. 4(*a*) shows the autocovariance relation for the spatial variation of $c_u$ in the vertical direction. The variance of $c_u$ is about 30 kPa$^2$, and, as is often observed, the covariance decays smoothly with increasing separation between the points. Fig. 4(*b*) shows the measurement error. The covariance is a spike at a separation of zero because measurement error is purely a local phenomenon. It includes random instrument or operator error and very small scale variations in soil properties. When the relations of Figs. 4(*a* and *b*) are added, the curve of Fig. 4(*c*) results. This is what would be expected from a set of field vane measurements. Fig. 4(*d*) shows the observed data.

Because the autocovariance functions for soil properties are different in the vertical and horizontal directions, the vertical autocorrelations should be done using values from the same boring, and the horizontal autocorrelations using data from different borings at the same elevation. Then each pair of data is used in (6) to compute the autocovariance as a function of the separation distance $r$. The results are then averaged or smoothed within each interval of $r$. Fig. 4(*d*) shows the results in the vertical direction for the marine clay at the B-6 site. The autocovariance is then extrapolated back to the vertical axis to obtain the autocovariance at $r = 0$. The difference between the observed variance and the extrapolated value must be random measurement error or small scale variations. The extrapolation can be done visually, or by fitting an appropriate curve, but in the latter case the apparent precision of the estimate may be misleading. De Groot and Baecher (1993) have published a more rigorous analysis of these data.

It must be emphasized that the purpose of the exercise is to separate the observed variation in measured soil properties into two categories. The first is the spatial variance in the field values, which should represent an effect that occurs at a scale that is relevant in the field and needs to be taken into account in the slope stability analysis. The second consists of random scatter due to error in measurements and very small scale fluctuations in soil properties. While these appear in the observed data, they do not affect the behavior of the soil that should be included in the stability analysis.

## Systematic Error

In addition to the data scatter and spatial variability, there are systematic errors in the estimation of the soil properties. The systematic error contributes to the analytical evaluation of slope stability very differently because the spatial variability averages over the volume of soil but the systematic error is consistent across the volume. Therefore, the systematic error propagates almost unchanged through an analysis and can have a much greater influence on the predictive uncertainty than does the spatial variability.

In the present example two factors contribute to the systematic error. First, a limited number of tests is used to determine the mean or trend of the shear strength. A smaller or larger set of measurements, or even a different set of measurements, would yield a different estimate. Standard



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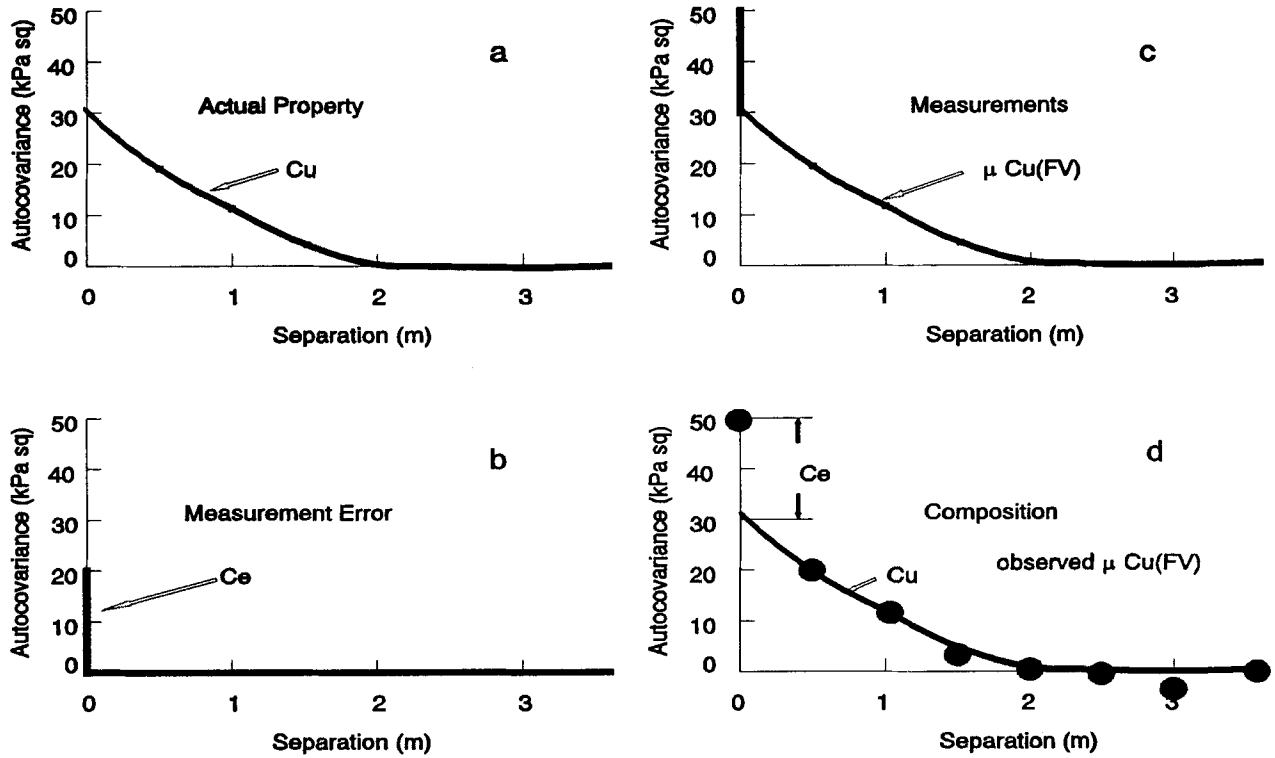



**FIG. 4. Composition of Autocovariance Function, Actual Properties (*a*) Combine with Measurement Error (*b*) to Give Total Distribution of Variances in Measurements (*c*); This Compares with Observed Data (*d*) (Assumes that $\mu = 1.0$ for this James Bay Marine Clay)**

statistical methods, such as those described by Benjamin and Cornell (1970), can provide confidence limits and variances of estimates obtained by regression analysis. In the present case of a mean value of a parameter $x$, the variance in the estimate of the mean from $n$ measurements is found from

$$V\{E[x]\} \approx V[x]/n \tag{15}$$

The second component of systematic error is bias. The experimental technique may not measure directly the quantity of interest or may measure it in some systematically erroneous way. The need to correct data from field vane tests is well known. The factor $\mu$ in (14) is Bjerrum's (1972) empirical correction factor as a function of the plasticity index, $I_p$. However, there is scatter in the data defining this relationship (Ladd et al. 1977; Azzouz et al. 1983). The scatter, which has a coefficient of variation of about 25%, represents the uncertainty in using this correction factor. On the basis of other observations on the marine clay from the present example, the mean of $\mu$ was taken to be 1.0, but, of course, this would not be true in most cases.

In separating spatial variability and systematic error it is perhaps easiest to think of spatial variation as describing the scatter of the soil property about the mean trend and of systematic error as uncertainty in the location of the mean trend itself. The only component of the former is inherent spatial variability after random measurement error has been removed, while the latter consists of measurement bias and statistical estimation error due to a finite number of measurements. The contribution of the spatial variability depends on the size of the region or failure surface over which it is effective in relation to the autocorrelation distance, $r_0$.

## APPLICATION TO EMBANKMENT DESIGN

Fig. 5 shows the cross section for the single- or multistage construction of dikes for the James Bay project (Ladd et al. 1983). The first alternatives were the construction of the embankment in a single stage, either to a height of 6 or 12 m with one berm. Undrained shear strength values were obtained from field vane tests, and the stability analyses were done by the simplified Bishop circular arc method of slices. The second alternative called for multistage construction. Following the installation of vertical drains, the first stage would consist of two berms built to a height of 12 m. Then the zone with the drains would be allowed to consolidate to 80% of primary consolidation, and the second stage would be built to a height of 23 m. In this case the undrained shear strength would be obtained from consolidated-undrained shear tests combined with knowledge of the initial and consolidated stress histories (Ladd 1991). The stability analysis was done by the Morgenstern-Price method of slices for a wedge-shaped failure pattern.

Fig. 6 is a simplified soil profile, which shows that the foundation is composed of 4 m of crust underlain by 8 m of a highly sensitive marine clay underlain in turn by 6.5 m of lacustrine clay, for a total thickness of clay of 18.5 m. It also shows the distribution of vane shear strength and the effective vertical stress history.

### Uncertainty in Soil Profile and Fill Properties

The single-stage cases involved many kilometers of embankment with a modest exploration program at any particular site along the embankment. The analysis had to include uncertainty about the thickness of the crust and



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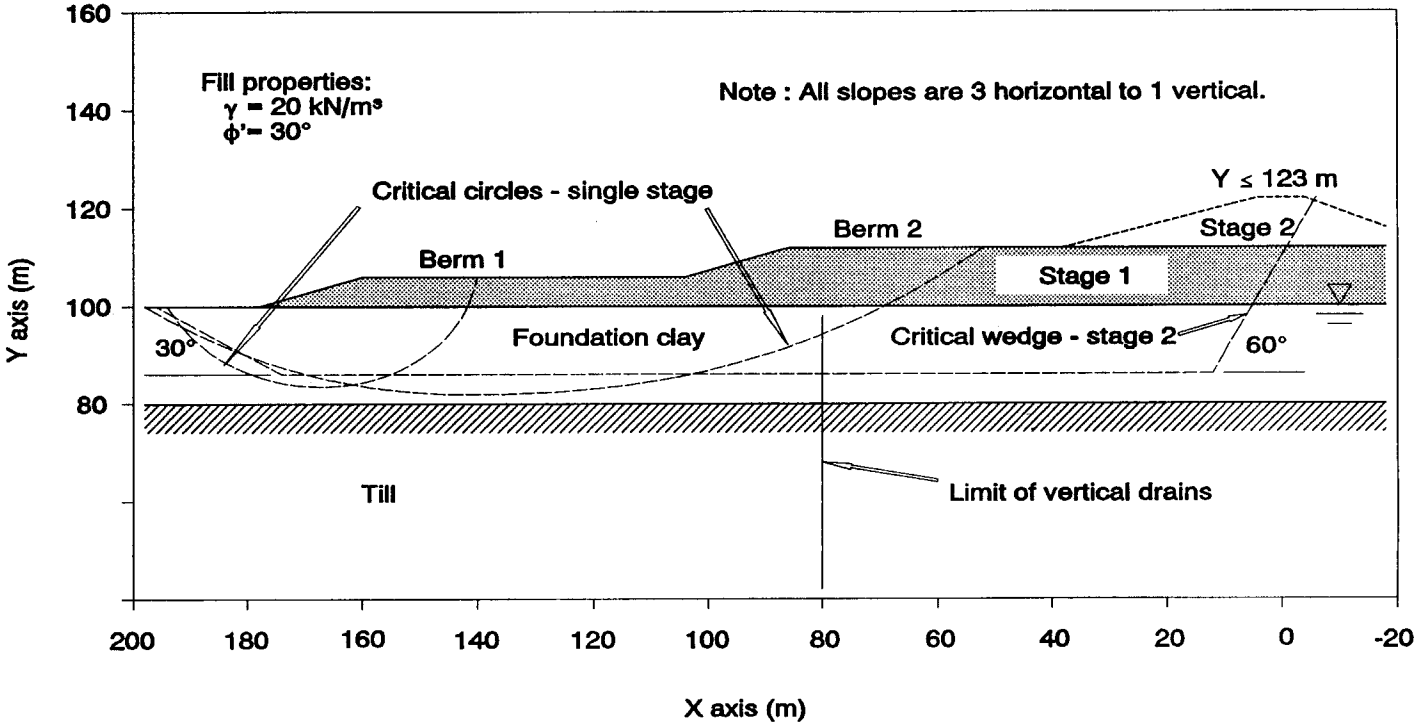



FIG. 5. Cross Section of SEBJ Design Problem (Ladd et al. 1983)

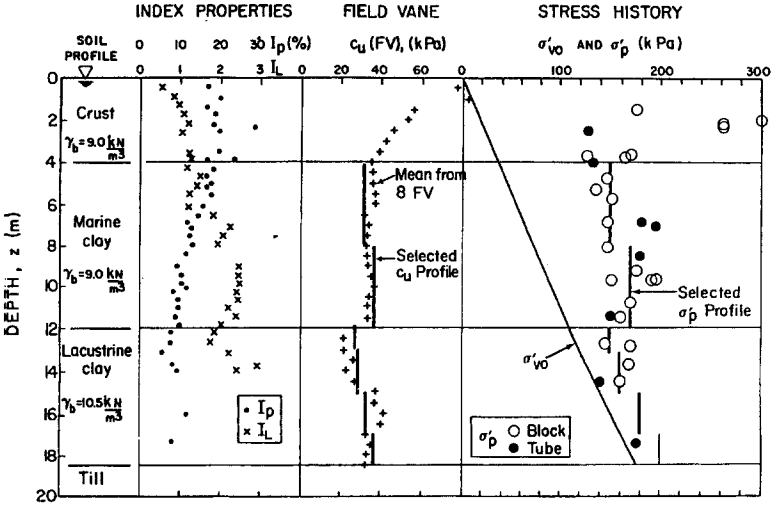



**FIG. 6. Soil Profile for Cross Section (Ladd et al. 1983)**

the depth to the till as well as the unit weight and friction angle of the fill. The multistage case involved a limited length of embankment with an extensive exploration and instrumentation program. Thus the geometry and properties of the foundation soils were better known.

The depth of the crust and the depth to the till were both treated as uncertain variables. From the results of field exploration, the uncertainties listed in Table 1 were established. The separation of the uncertainty into spatial and systematic components is important because averaging over the failure surface affects the contribution of spatial variability. The depth of the crust affects the undrained strength profile, and its spatial variance was estimated from the field vane data across the entire axis of the dike. The systematic statistical uncertainty for the vane soundings was estimated from (15) with $n = 27$.

Uncertainty in the depth to the till affects the location of the failure surfaces in the analysis. Since the critical surfaces are nearly tangent to the top of the till for $H = 12$ m, this uncertainty does not affect the average across the profile but is a systematic uncertainty even though the estimate of variance reflects spatial variance along the cross-section. That is, there is only one critical failure surface at any location. The location of this surface—and therefore its associated $F$—are dictated by the uncertain depth of the till at that location, but the uncertainty in the depth of the till comes about by the variation in depth over the whole project site. The estimate of systematic error reflects the effects of a finite number of borings.

Uncertainty in the density of the fill was taken to be composed equally of spatial and systematic components. This assumption and the estimated magnitudes of the variances were based on judgment. Selection of $\phi' = 30°$ reflected Canadian practice, and its variances were also based on judgment.

## Uncertainty in Shear Strength of Foundation Clay

### *Single Stage Cases*

Table 2 presents the uncertainty in $c_u$ used for the single stage analyses, based on the field vane data for the marine and lacustrine clays. The rel-



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atively small coefficient of variation selected for the bias in $\mu$ for the marine clay (Cov = 0.075) reflected detailed comparison with results from laboratory $CK_oU$ shear tests with different modes of failure, described by Lefebvre et al. (1988). This bias was doubled for the lacustrine clay, for which a similarly detailed comparison had not been made. The other components of spatial and systematic uncertainty were developed from the techniques described earlier. For example, the statistical coefficient of variation for the marine clay $\{[(0.236)^2/62]^{1/2} = 0.030\}$ follows from (15).

*Multistage Case*

For this case the shear strengths were established from a combination of undrained strength ratios and the in-situ stress history. For the overconsolidated intact clay beyond the limits of the vertical drains, no consolidation was assumed, and $c_u/\sigma'_p$ ratios were applied to the $\sigma'_p$ profile shown in Fig. 6. For the clay within the limits of the vertical drains, 80% consolidation was assumed under berm number 2, and the $c_u/\sigma'_{vc}$ ratios were applied to $\sigma'_{vc} = \sigma'_{vo} + 192$ kPa ($= 0.80 \times 12$ m $\times 20$ kN/m$^3$). Therefore, it is necessary to deal with uncertainty in both $c_u/\sigma'_p$ or $c_u/\sigma'_{vc}$ and $\sigma'_p$ or $\sigma'_{vc}$. Morgenstern-Price wedge stability analyses used anisotropic undrained strength profiles for compression, direct simple shear, and extension modes of failure.

The undrained shear strength ratios were obtained from $CK_oU$ triaxial compression and extension and Geonor direct simple shear tests run using the recompression technique and treated for strain compatibility to account for the effects of progressive failure (Ladd et al. 1983; Ladd 1991). Table 3 shows best estimates of these ratios. These were not intended to be conservative; otherwise, the reliability analysis would be inappropriately biased. In Table 3, the subscript $C$ indicates compression failure, $D$ indicates direct simple shear failure on a horizontal plane, and $E$ indicates extension failure.

Table 4 presents the uncertainties in the undrained strength ratios, the vertical preconsolidation and consolidation stresses, and the resulting values of undrained shear strength, $\tau$. Some uncertainties were developed by combining engineering judgment, experience, and experimental data. One of the major strengths of the reliability approach is that, instead of merely assuming conservative values of parameters, the engineer can use the best estimates and propagate the effects of the uncertainties in the analysis. The estimates are discussed separately in the following.

Undrained strength ratios: The coefficient of variation for the intact and consolidated clay were treated collectively for convenience. The spatial coefficient of variation was assumed equal to the average total scatter measured in $\tau/\sigma'$ for each of the three modes of failure. The systematic coefficient of variation had two components: one due to the statistical error calculated from (15), and the other the estimated bias. The latter assumed that the triaxial tests underestimated plane strain shear strengths by $10 \pm 5\%$ (Cov = 0.05) and that the strain rate used for the tests overestimated the in-situ strengths by $10 \pm 5\%$ (Cov = 0.05). Thus, these two errors cancel in the mean but cause a bias Cov $= \sqrt{2} \times 0.05 = 0.07$.

Preconsolidation pressures and consolidation stresses: Because very few oedometer tests were run (see Fig. 6), the spatial coefficient of variation of $\sigma'_p$ was assumed equal to that obtained from the field vane data, that is Cov = 0.18 and 0.27 from Table 2 for the marine and lacustrine clays, respectively. The systematic coefficients of variation reflected a combination of statistical error and judgment regarding the likely bias and were higher for the lacustrine clay because of the lack of data within the bottom half of the



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deposit (see Fig. 6). The vertical consolidation stress, $\sigma'_{vc}$, assumed no uncertainty in $\sigma'_{vo}$ and therefore reflected uncertainty in the average degree of consolidation, $U$, which would be estimated from field piezometer data, and the load imposed by berm 2. The values in Table 4 were all based on judgment and assumed the following: for spatial uncertainty, Cov[$U$] = 0.075 and Cov[load] = 0.05 to give Cov[$\sigma'_{vc}$] = 0.09; for systematic uncertainty, Cov[$U$] = 0.075 and Cov[load] = 0.0 to give Cov[$\sigma'_{vc}$] = 0.075, which was rounded to 0.08.

Undrained strength: The coefficient of variation for $\tau$ in Table 4 is obtained from

$$\text{Cov}^2[\tau] = \text{Cov}^2[\tau/\sigma'] + \text{Cov}^2[\sigma'] \tag{16}$$

The variance in $\tau$ actually used in the analysis then becomes

$$V[\tau] = \text{Cov}^2[\tau] \cdot E^2[\tau] \tag{17}$$

where $E[\tau]$ = mean undrained strength within the intact or consolidated layer of clay.

**Calculating Reliability Index**

It should be noted that the procedure minimizes $F$ over the range of failure surfaces and then calculates $\beta$. It does not minimize $\beta$ directly. The minimum $F$ surface and the minimum $\beta$ surface may not be identical. For example, Oka and Wu (1990) observed that, in a situation with two different but likely modes of failure, reliability calculations based on one mode only can be seriously in error. This is not the situation here.

*Single-Stage Embankments*

The results for the single stage dike built with one berm to a height of 12 m, without the model error, are displayed in Table 5. The circular arc analyses gave, for the critical failure circle, $E[F]$ = 1.453. The spatial variations tend to average out over the failure surface. By integrating the statistically varying shear strength over the failure surface it can be shown that the variance of the average shear strength due to spatial variability is

$$\frac{V[c_{u,\text{ave}}]}{V[c_u]} \approx \frac{2r_0}{L} \tag{18}$$

where $r_0$ = autocorrelation distance; and $L$ = length of the curve of the intersection of the failure surface with the plane of analysis. For the geometry of this case the reduction ratio becomes 0.2. This is applied to the spatial $V[F]$ of 0.0459.

For the case with a single stage to 6 m, $E[F]$ = 1.500, the critical failure circle is much smaller, and the reduction ratio for the spatial variation in (18) becomes 0.7. This leads to significantly larger variation in the computed factor of safety and a much lower value of $\beta$. (See Table 7 for a summary of these results.)

Fig. 7 shows graphically how the different parameters contribute to the variance in the factor of safety for the two single-stage cases. The bars represent the sum of the systematic uncertainty and the spatial uncertainty reduced due to averaging over the failure surface. The importance of the uncertainty in the strength of the lacustrine clay is obvious. The density of the fill and the strength of the marine clay are less important. The 6-m dike has a much smaller $\beta$ than the 12-m dike (1.84 versus 2.66) even though its



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**TABLE 7. Summary of Reliability Results and Effects of Model Error**

| Case (number of stages) (1) | $E[F]$[a] (2) | $V[F]$ Reduced spatial (3) | $V[F]$ Prior systematic (4) | $V[F]$ Model error[b] (5) | $V[F]$ Total (6) | $\beta = E[F] - 1/\sqrt{V[F]}$ (7) |
|---|---|---|---|---|---|---|
| Sgl., $H$ = 6 m | | | | | | |
| without model error | 1.500 | 0.050 | 0.024 | — | 0.074 | 1.84 |
| with model error | 1.575 | 0.050 | 0.024 | 0.012 | 0.086 | 1.96 |
| Sgl., $H$ = 12 m | | | | | | |
| without model error | 1.453 | 0.009 | 0.020 | — | 0.029 | 2.66 |
| with model error | 1.526 | 0.009 | 0.020 | 0.011 | 0.041 | 2.60 |
| Multi, $H$ = 23 m | | | | | | |
| without model error | 1.427 | 0.0012 | 0.0122 | — | 0.0134 | 3.69 |
| with model error | 1.498 | 0.0012 | 0.0122 | 0.0110 | 0.0244 | 3.19 |

[a] $E[F]$ with model error = $E[F]$ without model error × 1.05.
[b] $V[e]$ = (0.07 × increased $E[F]$)$^2$.

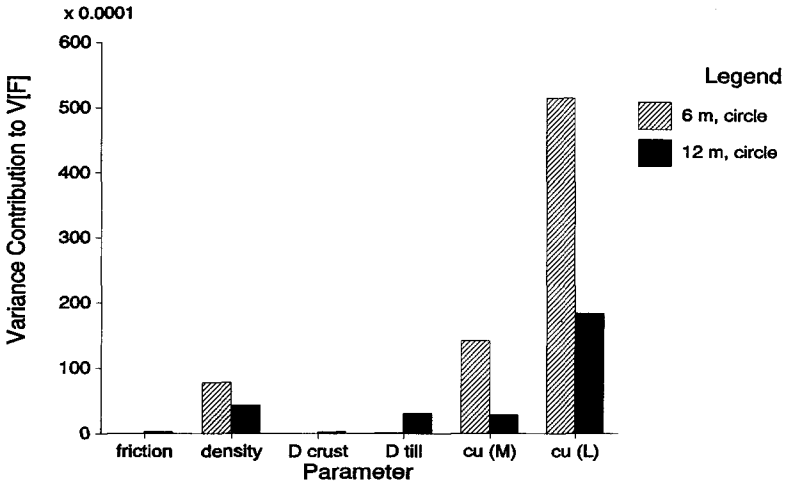



**FIG. 7. Variance Components for Factor of Safety of Single-Stage Dikes**

factor of safety is slightly higher (1.50 versus 1.45). The reduced reliability index reflects a somewhat larger systematic uncertainty and a much larger spatial uncertainty.

*Multistage Embankment*

Table 6 shows the results for the multistage dike. The failure surface in this case is large (see Fig. 5), but it is mostly horizontal so the appropriate autocorrelation distance for the shear strength is also large. The reduction factor for spatial variation of shear strength becomes 0.07. Because the overall uncertainty is much lower in this case, the reliability index, at 3.69, is the largest of all the cases. As in the single-stage cases, the strength of the lacustrine clay and the fill density are the most important contributors to the overall uncertainty. Since very little of the failure surface passes



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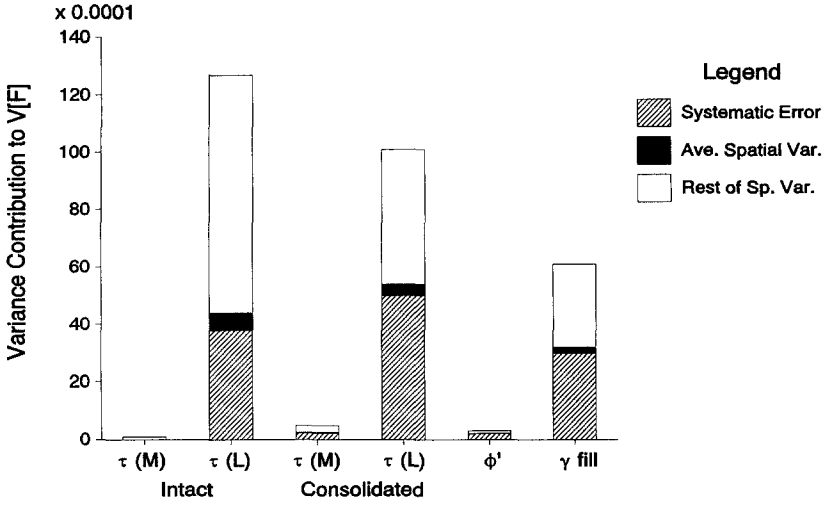



**FIG. 8. Variance Components for Factor of Safety of Multistage Dike**

through the marine clay, its uncertainty contributes little. The failure surface passes about equally through the intact and consolidated lacustrine clays, and each contributes about equally to the uncertainty. Although the coefficients of variation are larger for the intact clay, the mean strength is smaller. The contributions of the different parameters are illustrated in Fig. 8, which again shows the importance of the lacustrine clay. Fig. 8 shows the breakdown of the uncertainty between spatial and systematic contributions, and also shows the important effect of averaging the spatial uncertainty.

*Model Error*

Even if the soil parameters have been estimated correctly, the predictions of an engineering model can be expected to deviate from reality. Estimating the error introduced by this factor can be difficult; the most effective approach is to rely on empirical observations when they exist. An alternative is to use formal analysis. In the present case, three sources of model error were considered: three-dimensional failure compared to two-dimensional analysis (i.e., end effects), failure to find the most critical failure surface in the analysis, and numerical and rounding error. Azzouz et al. (1983) examined the contributions of end effects in eighteen case histories. They found that the ratio between the three-dimensional and the plane strain $F$ was 1.11 with a standard deviation of 0.06. These numbers were rounded to 1.1 and 0.05. Failure to find the lowest factor of safety is considered to overestimate $F$ by about 5% with a standard deviation of about 5%. The third factor was incorporated by assuming that it contributed 2% to the variability of the results. The combined effect of these three sources is to increase $E[F]$ by 5% and to create an added systematic uncertainty described by

$$\text{Cov} = (0.05^2 + 0.05^2 + 0.02^2)^{1/2} = 0.073 \approx 0.07 \qquad (19)$$



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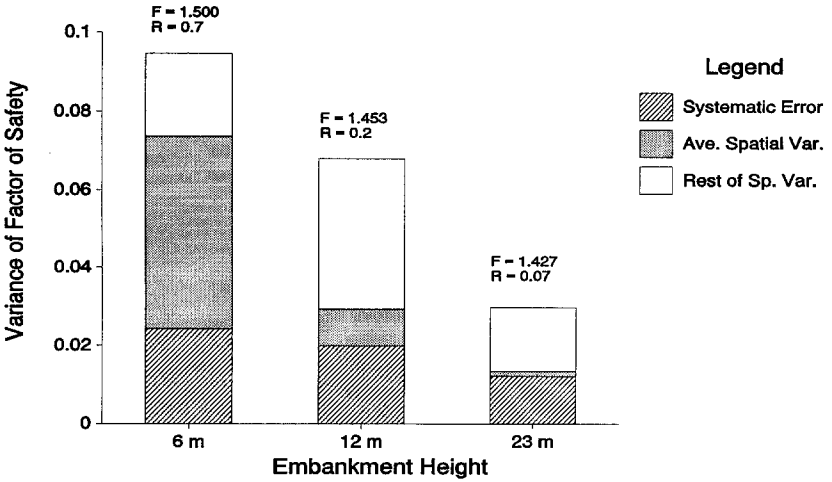



**FIG. 9. Separation of Spatial and Systematic Variance Components for Example Problems (Embankments of 6 m and 12 m Are Single Stage; 23-m Embankment Is Multistage)**

*Summary of Results for Three Cases*

The results for all three cases are summarized in Table 7. Although the factor of safety in the multistage case is slightly less than for the single-stage cases, the reliability index is significantly larger. Fig. 9 summarizes the overall contributions of systematic and spatial uncertainty. It shows the effect of the reduction in spatial uncertainty from averaging the soil parameters over the failure surface. Spatial variability becomes almost insignificant for the 23-m-high dike constructed in two stages. It should be noted that, in order to simplify calculations, the values of $R$ used for spatial averaging were based on analyses of the field vane data and therefore should apply only to the undrained strength for the single-stage cases with $H$ of 6 and 12 m and the part of the multistage case involving the intact clay. Different values of $R$ would exist for the fill density, friction angle, and normally consolidated undrained shear strength.

Table 7 shows that introducing model error can have unexpected results. For the single-stage dike with $H = 6$ m, $\beta$ increases from 1.84 to 1.96. There is little effect for the single-stage dike with $H = 12$ m; $\beta$ changes from 2.66 to 2.60. For the multistage dike $\beta$ decreases from 3.69 to 3.19. Thus, the computed reliability index can increase or decrease when the effects of model uncertainty are introduced. For a particular configuration and set of properties an increase in $E[F]$ might be offset by an increase in $\sigma[F]$.

## Extrapolation to Other Values of Parameters

The foregoing results follow from detailed consideration of a limited set of geometries and parameters. To develop consistent criteria for design, it is necessary to extrapolate them to other conditions. Two simple possibilities exist. On the one hand, one could assume that for a basic geometry the variance or standard deviation of $F$ is constant and apply (8). On the other hand, one could assume the coefficient of variation is constant and apply (9). The consequences of the two assumptions are shown in Fig. 10. Constant



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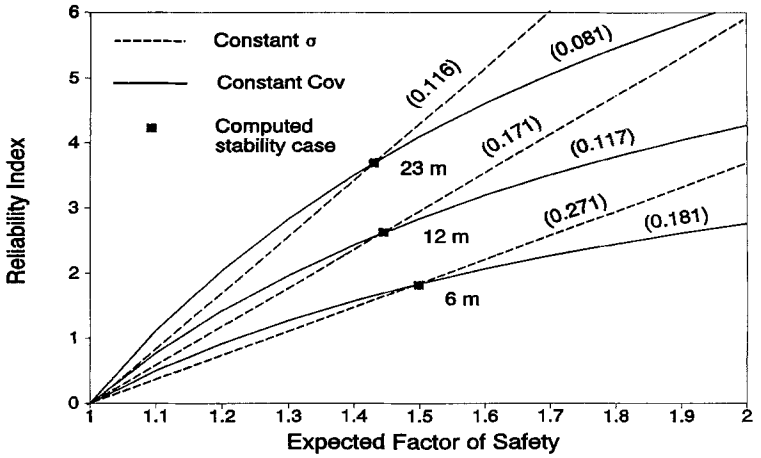



**FIG. 10. Projection of Reliability Index versus Factor of Safety (without Model Error)**

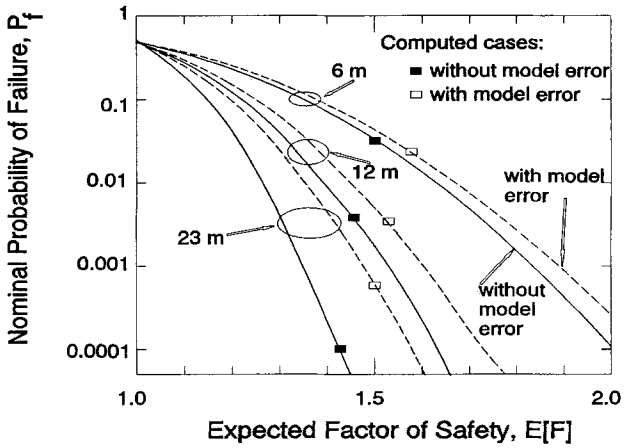



**FIG. 11. Nominal Probability of Failure versus Computed Factor of Safety for Example Problems**

variance means that the absolute uncertainty in $F$ does not change with the target value of $E[F]$. Constant coefficient of variation means that the uncertainty in $F$ changes in proportion to $E[F]$. Since the derivatives (or in this case the divided differences $\Delta F/\Delta x_i$) are for the most part nearly constant for different values of $F$, a projection based on constant variance seems the better choice. It is also computationally simpler.

Fig. 11 shows the projected nominal probabilities of failure as functions of the computed factors of safety for the example stability problems. The probabilities are computed on the assumption that $F$ is normally distributed, leading to the relation plotted in Fig. 1. Until now the form of the distribution of $F$ has not entered into these calculations. Curves are plotted for results both with and without the effects of model uncertainty. From Fig. 11 the



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designer can determine the computed factors of safety that correspond to a desired target nominal probability of failure.

*Effects of Types of Failure*

The uncertainties associated with the three design cases differ from one another because each reflects a different mixture of spatial variability and systematic uncertainty. This has an important implication. A 1% probability of failure deriving from spatial variability alone implies that, on average, one per cent of the length of a long dike would fail. Conversely, a 1% probability of failure due to systematic uncertainty alone would imply that, on average, one out of 100 of a large number of similar dikes would fail along its entire length. In any specific case, such as this one, the probability of failure derives from both spatial variability and systematic uncertainty.

Fig. 9 shows the contributions of the two components for the three cases. In the single-stage case with $H = 6$ m the distribution of $F$ for different sections along the dike is broad, but the uncertainty on the location of the mean is small. In the multistage case with $H = 23$ m, the distribution of $F$ along the dike is narrow, but there is proportionally more uncertainty on the location of the mean. In other words, the uncertainty in the case of a single-stage dike with $H = 6$ m deals principally with the fraction of the dike that will fail, but the uncertainty in the case of a multistage dike deals principally with the failure of all or a large portion of the dike.

*Target Probability of Failure*

As mentioned earlier, Baecher et al. (1980) determined that the historical rate of failure of dams is about $10^{-4}$ per dam-year. The details of failures are often imprecise in the catalogs from which this result was computed, but it appears that about one-third of the failures were due to overtopping and another one-third to internal erosion, piping, or seepage. These numbers are close to what would be expected from professional judgment and experience. The remaining one-third are due to slides and other mechanisms, which include the subject of the present analysis.

The relative contribution of different modes of failure should be kept in mind when a target probability of failure is being selected. It makes little sense to reduce the computed probability of failure due to slope stability problems if other causes of failure are not addressed at the same time. The historical data would suggest that, for a dam with a life of 100 years, the overall probability of failure is 0.01, in the absence of specific attention during design and construction to mitigate the hazards. Since well less than one third of the failures are due to slope instability per se, a target probability of failure of slope stability failure of about 0.001 seems reasonable for design purposes and consistent with past experience. Fig. 11 indicates that to achieve this target probability the stability analysis should give factors of safety of 1.84, 1.53, and 1.36 for the 6-, 12-, and 23-m cases, respectively. If the effects of model uncertainty were included, the values of desired factor of safety would increase to 1.91, 1.63, and 1.48. However, this effect was ignored in further results.

*Consistent Factors of Safety*

The present analysis was concerned with stability during construction due to failure of the foundation clay. The consequences of such a failure can be expected to consist primarily of the costs of reconstruction. Failure of the embankment after the facility is in operation would involve much larger



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**TABLE 8. Consistent Factors of Safety (Without Model Error)**

| Case (1) | Target $P_f$ (2) | Consistent factor of safety (3) |
|---|---|---|
| Single stage, $H$ = 6 m, circular failure surface | 0.01 | 1.63 |
| Single stage, $H$ = 12 m, circular failure surface | 0.001 | 1.53 |
| Multistage, $H$ = 23 m, wedge failure surface | 0.0001 | 1.43 |

economic costs as well as other public consequences. This means that different target reliability indices or probabilities of failure are appropriate for different stages in the life of the project. The selection of such values is a difficult task for owners, designers, and public officials. Even for the present problem, the consequences of failure are quite different in the different cases. Because experience with staged construction is less fully evaluated and the consequences of failure of a major portion of the completed multistage embankment are expected to be greater than those of the failure of the 12-m-high embankment, the target probability for the multistage case might be reduced to 0.0001, implying a calculated factor of safety of about 1.43. Conversely, the consequences of failure of a portion of the smallest dike might be only 1/10 as great. The target probability then becomes 0.01 and the desired factor of safety about 1.63. Based on the revised target probabilities, one obtains the consistent, desired factors of safety shown in Table 8. The range in factors of safety for the different cases is now much smaller than for the same $P_f = 0.001$. These results led to the recommendation that $F = 1.50$ be adopted for feasibility studies and preliminary cost estimates for both single and multistage dikes. It is interesting to note that a prior committee had recommended using $F = 1.4$ for minor dikes and $F = 1.7$ for major (i.e. multistage) dikes.

## WHITMAN'S OBSERVATIONS ON THE USE OF PROBABILISTIC METHODS

In his Terzaghi Lecture, Whitman (1984) gave an excellent summary of the state of the art of probabilistic applications in geotechnical engineering. He observed

> The spectrum of slope stability problems is quite broad. Toward one extreme are situations for which there is already extensive practical experience and for which there is considerable agreement as to an appropriate safety factor to use for design. On the other hand, there are problems involving soils for which there is very little direct experience. In this case, there may be considerable doubt as to the proper method for computing safety, and the probability of failure is rather high. Some of the problems of stability in very soft clays . . . fall in this category, as do many slopes in residual soils. In both these situations, reliability theory is of little use in the selection of an appropriate safety factor. Many problems fall into one or the other of these categories, which may explain why there are few, if any, papers in the literature describing the actual application of reliability theory in design decisions concerning a slope.



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It is hoped that the present paper provides a partial remedy to the last observation.

Whitman also provided an excellent explanation of the use of event trees for the evaluation of overall risk of geotechnical structures, together with hypothetical examples of well and poorly designed dams. An analysis of this sort must be done if the engineer is to evaluate the overall probability of failure of an engineered structure. Probabilistic risk assessments have been performed for a variety of critical facilities, such as nuclear plants. However, as Whitman demonstrated, much judgment goes into assessing the probabilities associated with some events; when a consensus about the appropriate values cannot be reached, the analysis breaks down.

## SUMMARY AND CONCLUSIONS

### General Conclusions

The review of reliability analysis and its application to the problems of the embankments at the James Bay project leads to several major conclusions. These include the following:

1. The reliability index $\beta$, as defined in (8), provides a more meaningful measure of stability than the factor of safety $F$. Geotechnical engineers have long recognized that $F$ has little physical meaning and that the choice of a satisfactory value is fraught with difficulty. In contrast, the reliability index describes safety by the number of standard deviations (i.e., the amount of uncertainty in the calculated value of $F$) separating the best estimate of $F$ from its defined failure value of 1.0. It is a more explicit approach.
2. The uncertainty in the values of the soil properties is a major contributor to the uncertainty in the stability of slopes and embankments. It consists of two portions: scatter and systematic error. The first comprises real spatial variability and random measurement error. These represent variations about the trend of the parameters. The systematic error includes both a statistical uncertainty in the expected values and the effects of bias. The latter is much the more difficult to evaluate.
3. The structure of the spatial variation can be used to estimate the level of random noise in soil property data and to eliminate it from the calculation of reliability index. The effects of the spatial variability on the computed reliability index are further reduced because the variability is averaged over a region or failure surface and it is only its average contribution to the uncertainty that is of interest.
4. Several difficulties arise in practical assessment of scatter and systematic error. There are often not enough data to evaluate the autocorrelation distance statistically. Evaluating the effects of bias introduces many additional concerns. Bias is often very important, and the associated uncertainty can be a significant contributor to the overall uncertainty. Since bias is often ignored in theoretical treatments of analytical procedures and it is difficult to quantify, the engineer must often rely on judgment to establish its contribution. The strength of the reliability analysis is not that one can get a better estimate of each of these uncertainties but that one can deal with them explicitly and coherently.
5. Uncertainties in soil properties yield a lower bound estimate of the probability of failure, not the absolute probability of failure. That requires a more elaborate probabilistic risk analysis involving fault trees or other



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method of evaluating risk due to all contingencies (Whitman 1984). However, for most practical applications the calculation of relative probability of failure is sufficient for parametric analysis.

6. A 1% chance that the whole slope will fail is different from a near certainty that one per cent of a long embankment will fail, and reliability analysis allows such a distinction.

## Analysis of James Bay Embankments

The analysis of the James Bay embankments illustrates a procedure for evaluating the reliability index. First, the uncertainties are identified and the relevant statistical properties are estimated. Second, the partial derivatives of $F$ with respect to the uncertain variables are calculated. Third, the mean and variance of $F$ are calculated and the resulting $\beta$ evaluated.

Once the uncertainties in the soil properties, geometry, loads, and other contributing factors have been established, it is necessary to propagate their effects through the stability analysis. Eqs. (11) and (13) provide one way to do so. The first-order, second-moment approach is a very powerful approximate method to deal with the calculations. The Taylor series combined with a divided difference approximation to the derivatives has proven very effective.

In the analysis the effects of both spatial and systematic uncertainty are considered. The averaging effect tends to reduce greatly the contribution of the spatial uncertainty. In the case of the multistage construction spatial uncertainty contributes so little that it could almost be ignored without affecting the results. The small embankment with a short failure surface has a large uncertainty in $F$ due to spatial variation in undrained strength. This means that several small failures along the length of the embankment are likely. The multistage embankment with a long failure surface has almost all the uncertainty in systematic error, which leads to a small probability of failure along a long section of the embankment. When the autocorrelation distance is not known, the analysis can at least give point estimates of the reliability index without providing information on the length of embankment likely to fail.

The example also illustrates the contribution of model error, and the difficulty in determining the magnitude of such bias effects. These effects have not always been included in reliability analyses.

However the solution is carried out, it is important to deal rationally with the spatial variability of the parameters. In particular, the averaging effect on variations that are due to inherent spatial scatter and are not systematic must be accounted for or the resulting probabilities of failure will be too high. The technique used here is one appropriate method. More formal, detailed methods from random field theory can also be used, but they have found relatively little practical employment because of the large amount of data needed to use them properly.

## Recommendations

A major use of reliability methods is to establish consistent targets for the design of an embankment. These are usually expressed in terms of the desired factor of safety. In the present example the analysis was directed at stability during construction and not after impounding water, but the same techniques could be used in that case as well. The desired factor of safety is established by balancing the costs of achieving a degree of safety with the costs associated with failure, multiplied by the probability of failure.



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Reliability approaches to the evaluation of the stability of slopes and embankments have evolved to the point that they should be applied more often in practice. Since the analysis requires considerably more effort than does a simple deterministic analysis, the engineer must have a good reason to use the reliability approach. Among the best reasons are accounting rationally for the propagation of uncertainty in the analysis, not relying exclusively on an arbitrary factor of safety, and dealing coherently with the interaction of several effects and their relative contributions.

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## APPENDIX II. NOTATION

*The following symbols are used in this paper:*

$C[\ ,\ ]$ = covariance of two variates enclosed in brackets;
$C_x(r)$ = autocovariance over distance $r$;
Cov[ ] = coefficient of variation of variate enclosed in brackets;
$c_e$ = random experimental error;
$c_u$ = undrained shear strength;
$c_u(FV)$ = undrained shear strength from field vane test;
$D_{\text{crust}}$ = thickness of crust;
$D_{\text{till}}$ = depth to till;
$E[\ ]$ = expected value of variate enclosed in brackets;
$e$ = random error;
$F$ = factor of safety;
$g(x)$ = arbitrary function of $x$;
$H$ = height of embankment;
$L$ = length over which spatial variation is averaged;
$(L)$ = indicates value for lacustrine clay;
$(M)$ = indicates value for marine clay;
$n$ = number of data points or tests;
$P_f$ = probability of failure;
$R$ = reduction factor to account for averaging of spatial variation;
$R_x(r)$ = autocorrelation coefficient over distance $r$;
$r$ = distance for correlation;
$r_0$ = autocorrelation distance;
$V[\ ]$ = variance of variate enclosed in brackets;
$x$ = variate or variable;
$x_i$ = value of $x$ at particular point;
$y$ = variate or variable;
$\beta$ = reliability index;
$\gamma$ = unit weight;
$\mu$ = correction coefficient for field vane data;
$\sigma[\ ]$ = standard deviation of variate in brackets;



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$\sigma'_p$ = effective preconsolidation pressure;
$\sigma'_{vc}$ = effective vertical consolidation stress;
$\sigma'_{vo}$ = effective initial vertical stress;
$\tau$ = shear strength; subscripts $C$, $D$, and $E$ indicate triaxial compression, direct simple shear, and triaxial tension, respectively; and
$\phi'$ = effective friction angle.

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